

Yu-Wei Fan

Education

Harvard University

Ph.D., Mathematics, 2014-2019

Taiwan University

B.S., Mathematics, 2008-2012

Employment

Fudan University and SIMIS (joint appointment)

Associate Professor (tenured), 2025.09-present

Tsinghua University

Assistant Professor (tenure track), 2021.12-2025.08

University of California, Berkeley

Morrey Visiting Assistant Professor, 2019.08-2021.11

Research Interests

Algebraic Geometry, Dynamical Systems, Symplectic Geometry

Publications and Preprints

- (1) A space of stability conditions that is not a length space.
Science China Mathematics, accepted.
- (2) Fourier-Mukai numbers of K3 categories of very general special cubic fourfolds.
Israel Journal of Mathematics, Vol. 268 (2025), pp. 427–437, with K.-W. Lai.
- (3) Stokes matrices and exceptional isomorphisms.
Mathematische Annalen, Vol. 390 (2024), pp. 4041–4086, with J. P. Whang.
- (4) Shifting numbers of abelian varieties via bounded t-structures.
manuscripta mathematica, Vol. 174 (2024), pp. 775–784.
- (5) Asymptotic shifting numbers in triangulated categories.
Advances in Mathematics, Vol. 428 (2023), 109163, with S. Filip.
- (6) New rational cubic fourfolds arising from Cremona transformations.
Algebraic Geometry, Vol. 10, No. 4 (2023), pp. 432–460. with K-W. Lai.
- (7) Contractibility of space of stability conditions on the projective plane via global dimension function.
Mathematical Research Letters, Vol. 30, No. 1 (2023), pp. 51–87. with C. Li, W. Liu, and Y. Qiu.
- (8) Counting special Lagrangian classes and semistable Mukai vectors for K3 surfaces.
Geometriae Dedicata, Vol. 217, Issue 5 (2023), 89, with J. S. Athreya and H. Lee.
- (9) Attractor mechanisms of moduli spaces of Calabi-Yau 3-folds.
Journal of Geometry and Physics, Vol. 185 (2023), 104724, with A. Kanazawa.

- (10) Systolic inequalities for K3 surfaces via stability conditions.
Mathematische Zeitschrift, Vol. 300 (2022), pp. 301–323.
- (11) On pseudo-Anosov autoequivalences.
Advances in Mathematics, Vol. 384 (2021), 107732, with S. Filip, F. Haiden, L. Katzarkov, and Y. Liu.
- (12) Categorical polynomial entropy.
Advances in Mathematics, Vol. 383 (2021), 107655, with L. Fu and G. Ouchi.
- (13) Weil–Petersson geometry on the space of Bridgeland stability conditions.
Communications in Analysis and Geometry, Vol. 29, No. 3 (2021), pp. 681–706, with A. Kanazawa and S.-T. Yau.
- (14) Mirror of Atiyah flop in symplectic geometry and stability conditions.
Advances in Theoretical and Mathematical Physics, Vol. 22, No. 5 (2018), pp. 1149–1207, with H. Hong, S.-C. Lau, and S.-T. Yau.
- (15) Entropy of an autoequivalence on Calabi–Yau manifolds.
Mathematical Research Letters, Vol. 25, No. 2 (2018), pp. 509–519.
- (16) An invitation to categorical dynamical systems.
Ergodic Theory and Dynamical Systems, Proceedings of the Workshops University of North Carolina at Chapel Hill 2021.
- (17) Stokes matrices, character varieties, and braids.
Proceedings of the 2020 ICCM Annual Meeting.
- (18) Notes on the deformed Hermitian–Yang–Mills equations and the large scaling limits of stability conditions.
Preprint.
- (19) Special Lagrangians and Bridgeland stable objects beyond geometric stability conditions: the product case.
Submitted.
- (20) Mirror symmetry for lattice-polarized abelian surfaces.
Submitted, with K.-W. Lai.
- (21) Nielsen realization problem for Bridgeland stability conditions on generic K3 surfaces.
Submitted, with K.-W. Lai.
- (22) On entropy of $\mathbb{P}$ -twists.
Submitted.

Selected Conference/Workshop Talks

- China–Latin America Conference in Mathematics and Applications, SIMIS, Shanghai, March 2026
- The 2025 Annual International Congress of Chinese Mathematicians, Shanghai, January 2026
- International Conference on Symplectic Dynamics, Southern University of Science and Technology, August 2025
- Hefei Conference on Algebraic Geometry and Arithmetic Geometry, University of Science and Technology of China, May 2025

- Conference on advanced developments in symplectic topology and symplectic dynamics, Tianyuan Mathematical Research Center, April 2025
- The 2024 Annual International Congress of Chinese Mathematicians, Shanghai, January 2025
- HSE/BIMSA conference on Algebraic Geometry and Mathematical Physics, November 2024
- Workshop on Calabi–Yau geometry, ShanghaiTech University, October 2024
- Conference on Geometry and Moduli, Zhejiang University, July 2024
- The 2023 Annual International Congress of Chinese Mathematicians, Shanghai, January 2024
- Algebraic Geometry Workshop, University of Science and Technology of China, November 2023
- International Congress of Basic Science, Beijing, July 2023
- Conference on Algebraic Geometry and Related Topics, Nanjing University, May 2023
- Homological Algebra of the Infrared, TSIMF Sanya, January 2023
- AMS Spring Eastern Sectional Meeting (Special Session on Mirror Symmetry and Enumerative Geometry), Brown University, March 2021
- ICCM annual meeting, University of Science and Technology of China, December 2020
- Hodge Theory and Rationality, Institute of the Mathematical Sciences of the Americas, Miami, October 2020
- Workshop on Birational Geometry, Institute of Mathematics and Informatics, Bulgarian Academy of Sciences, June 2020 (*canceled due to COVID-19*)
- Moduli on Chestnut Hill, Boston College, March 2020
- Homological Mirror Symmetry and Topological Recursion, Miami, January 2020
- Homological Mirror Symmetry and Self Organized Criticalities, Institute of the Mathematical Sciences of the Americas, Miami, September 2019
- Aspects of Mirror Symmetry, Chiba University, July 2019
- PIMS Postdoctoral Training School on Stochastic Dynamics and Hodge Theory, University of Alberta, March 2019
- Holomorphic Differentials in Mathematics and Physics, Simons Center, February 2019
- AMS Grad Student Conference in geometry and topology, Brown University, April 2018
- BU–Keio Workshop, Boston University, June 2017
- RTG/FRG Mirror Symmetry Conference for Young Researchers, University of Michigan, May 2017

Selected Seminar Talks

- ZUMA Geometry and Topology Seminar, Zhejiang University, May 2026
- Sun Yat-Sen University, May 2026
- SISSA/IGAP Complex Geometry Seminar, March 2026
- BIMSA, Disquisitions on Monoidal Categories and Operads, March 2026

- BIMSA AG Seminar, January 2025
- Southern University of Science and Technology, August 2024
- Morningside Center, Symplectic Geometry and Mathematical Physics Seminar, June 2024
- The Greater Bay Area Geometry and Mathematical Physics Seminar, April 2024
- Northeast Normal University, October 2022
- Capital Normal University, April 2022
- New Directions in Group Theory and Triangulated Categories, November 2021
- Geometric Group Theory in East Asia Webinar, October 2021
- IBS Center for Geometry and Physics, Symplectic Monday, June 2021
- Zhejiang University, June 2021
- University of British Columbia, Algebra and Algebraic Geometry seminar, February 2021
- Algebraic Geometry in East Asia Seminar, December 2020
- Symplectic Zoominar (CRM-Montreal, Princeton/IAS, Tel Aviv, Paris), November 2020
- UC Irvine, Differential Geometry Seminar, October 2020
- Rutgers University–Newark, Colloquium, October 2020
- National University of Singapore, Topology and Geometry Seminar, September 2020
- ICCM Math Physics Seminar, July 2020
- UC Santa Barbara, Seminar on Geometry and Arithmetic, May 2020
- Harvard University, CMSA Mathematical Physics Seminar, April 2020
- Kyoto University, Algebraic Geometry Seminar, January 2020
- Southeast University, Nanjing, China, December 2019
- UC Berkeley, Northern California Symplectic Geometry Seminar, November 2019
- MSRI, Learning seminar on stability conditions, November 2019
- National Taiwan University, NCTS Seminar in Algebraic Geometry, June 2019
- UMass Amherst, Valley Geometry Seminar, April 2019
- Harvard University, Differential Geometry Seminar, April 2019
- Boston College, NT & AG Seminar, January 2019
- Harvard University, Student/Postdoc Symplectic Geometry Seminar, October 2018
- Columbia University, Symplectic Geometry, Gauge Theory, and Categorification Seminar, September 2018
- Tsinghua University, Beijing, Geometric Representation Seminar, June 2018
- Kavli IPMU, Mathematics-String Seminar, May 2018
- TIMS, Taiwan, Algebraic Geometry Seminar, May 2018
- Brown University, Algebra and Algebraic Geometry Seminar, May 2018
- Kansas State University, M-seminar, April 2018
- Harvard University, Mirror Symmetry Seminar, March 2018

- Boston University, Geometry and Physics Seminar, February 2018
- University of Missouri, Geometry and Topology Seminar, November 2017
- Harvard University, CMSA Algebraic Geometry Seminar, September 2017
- Harvard University, CMSA Mathematical Physics Seminar, September 2017
- IHES, Séminaire Mathjeunes, June 2017
- University of Vienna, June 2017

Teaching

At Fudan University and SIMIS:

- Spring 2026: *Riemann Surfaces*
- Fall 2025: *Algebraic Varieties and Their Derived Categories*

At Tsinghua University:

- Spring 2025: *Algebra 2*
- Fall 2024: *Topics in Derived Categories*
- Spring 2024: *Algebra 0*
- Fall 2023: *Ordinary Differential Equations*
- Spring 2023: *Mathematics from examples*
- Fall 2022: *Geometry and Symmetry*
- Spring 2022: *Categories and dynamical systems*

At UC Berkeley:

- Fall 2021: *Honors Linear Algebra and Differential Equations*
- Spring 2021: *Introduction to Analysis* and *Introduction to Complex Analysis*
- Fall 2020: *Honors Linear Algebra and Differential Equations*
- Spring 2020: *Introduction to Analysis*
- Fall 2019: *Honors Linear Algebra and Differential Equations* and *Introduction to Analysis*

At Harvard University:

- Spring 2019: *Theoretical multivariable calculus and real analysis*, CA
- Fall 2018: *Symplectic Manifolds and Lagrangian Submanifolds*, CA
- Fall 2017: *Multivariable Calculus*
- Fall 2016: *Multivariable Calculus*, Coach

Mentoring

Undergraduate thesis mentored:

- Jie Fu (2025), *Quantum Cohomology and Paths in Stability Spaces*

Service

Seminars (co-)organized:

- SIMIS Algebraic Geometry Seminar, 2025–present
- YMSC–BIMSA Geometry and Dynamics Seminar, 2023–2025
- YMSC Seminar on Algebraic and Complex Dynamics, 2022-2023

Conferences (co-)organized:

- Workshop on Mirror Symmetry and Stability, Harvard University, March 2019

Reviewing:

- Referee for articles submitted to: *Duke Mathematical Journal*, *Journal of Differential Geometry*, *Compositio Mathematica*, *Advances in Mathematics*, *Algebraic Geometry*, *International Mathematics Research Notices*, *Journal of the London Mathematical Society*, *Science China Mathematics*, *Bulletin of the London Mathematical Society*, *Mathematical Research Letters*, *Journal of the Mathematical Society of Japan*, *manuscripta mathematica*, *Geometriae Dedicata*, *Experimental Mathematics*, *Contributions to Algebra and Geometry*
- Referee for proposals submitted to: *U.S.-Israel Binational Science Foundation*